NUS National University of Singapore

10 pages

ANSWER BOOK

T-10

Student ID A0172562E

Module Code / Title CS3230 / MidTerm 2

Semester / Academic Year 1 / AY1920 Number of books handed in 1

INSTRUCTIONS TO CANDIDATES

1. Write your answers on the single line pages only.
2. The blank pages may be used for rough work.
3. Begin each answer on a fresh page.
4. Write the question number at the top of each page.
5. Write the question numbers (in the order the questions were answered) in the relevant column in the table on this page.
6. Do not tear out any pages from the answer book. Fasten all answer books and graph paper (if any) together.
7. Do not remove any answer books, used or unused, from the examination room.

Question	EXAMINER'S USE ONLY		
	Marking		
	Int.	Ext.	Aver.
2-1	3		
2-2	20		
2-3	2		
Total	25		

Question 2-1

Write answers on this side of the paper only.

Do not write on either margin

For this problem, we will use the Bipartite Matching algorithm covered during lecture.

First, ~~the~~ I create a Bipartite Graph with 2 disjoint and independent sets of ~~n^2 vertices~~ the undeleted squares. We do so by iterating through all n^2 tiles on the checkboard and check if deleted[i,j] is true for tile i,j. If it is TRUE, we add ~~the~~ [i,j] into the set. We replicate this for the other set in the Bipartite Graph. This will take $O(n^2)$ time.

Then, we create an edge between 2 vertices in different sets if the 2 vertices are adjacent, which is only ~~these~~ in 4 cases, ~~if~~ given vertex [i,j] and vertex [k,l]. ~~i~~

Add an edge if $\pm(i-k) = 1$ and $\pm(j-l) = 1$

This ~~will take~~ $O(n^4)$ time as we will have to iterate through both sets of ~~$O(n^2)$~~ $O(n^2)$ vertices. ~~to see the~~ if

Next, we check for the number of undeleted tiles by iterating through ~~all the~~ the deleted[i,j] for all tiles [i,j]. This will take $O(n^2)$ time.

Lastly, we will use the algorithm covered in lecture of Bipartite Matching to compute the maximum matching of the graph. If the maximum matching is equal to the number of undeleted tiles in the checkerboard, we output TRUE. Else, we output FALSE. The algorithm takes $O(VE)$ $= O(n^2 \times 4n^2) = O(n^4)$ time.

Hence, the overall time complexity is $O(n^4)$.

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